Quantum Entanglement, Part 3

Lecture 9: Friedmann Dynamics, Matter, Radiation, and Vacuum Energy

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Chapter 1

Friedmann Dynamics: Matter, Radiation, and Vacuum Energy

Overview. The kinematics of an expanding universe can be stated in one line: physical separations grow with a scale factor $a(t)$. The problem now is to determine that function. We shall recover the flat-space Friedmann equation from Newtonian energy conservation, solve it first for ordinary matter, and then see why radiation and vacuum energy lead to entirely different cosmic histories.

1.1 The geometry to be explained

At a fixed cosmic time, the large-scale universe appears homogeneous and isotropic. Galaxies, clusters, and superclusters are certainly real features, but on scales of hundreds of millions or billions of light-years their distribution becomes remarkably uniform. This empirical regularity is called the cosmological principle. The name does not make it a theorem; the force of the principle comes from observation.

There is a second empirical fact. Within present accuracy, large spatial triangles behave as Euclidean triangles. The comparison with surveying a football field is useful. If tightly stretched lines make triangles whose angles add to 180°, one has not proved that Earth is globally flat. One has shown that Earth's curvature radius is large compared with the field. Cosmic surveying has the same logical form: space may have a very large curvature radius, but over the observable region it is well approximated as flat.

The reference geometry is Minkowski spacetime,

$$d\tau^2 = dt^2 - \left(dx^2 + dy^2 + dz^2\right). \tag{1.1}$$

The bracketed expression is ordinary three-dimensional Pythagoras. In general relativity, by contrast, the coefficients of the coordinate differentials are fields: they may depend on position and time, and mixed terms may also occur. Those metric coefficients are the relativistic successors of the Newtonian gravitational field.

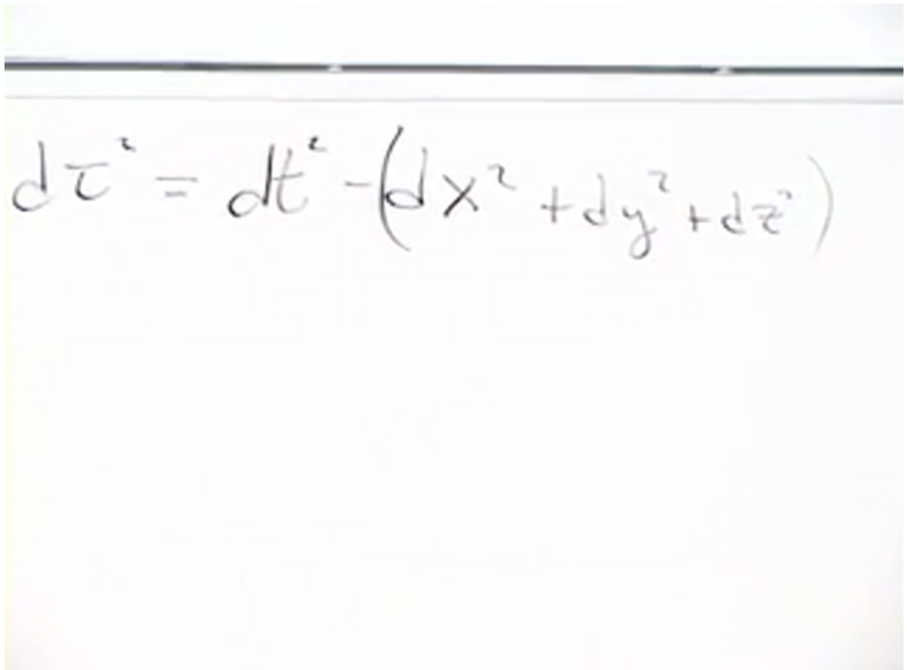

Figure 1.1: The Minkowski proper-time element written at the beginning of the lecture. Source frame at 00:01:50.

For a spatially flat expanding universe, homogeneity and isotropy leave one time-dependent scale factor:

$$d\tau^2 = dt^2 - a^2(t)\left(dx^2 + dy^2 + dz^2\right). \tag{1.2}$$

We use units in which $c = 1$, so time and length have the same dimensions. There are two consistent conventions. One may give x, y, z dimensions of length and take a dimensionless, or treat x, y, z as dimensionless labels and let a carry the dimensions of length. The lecture uses the second convention because it makes the physical meaning of $a(t)$ visible.

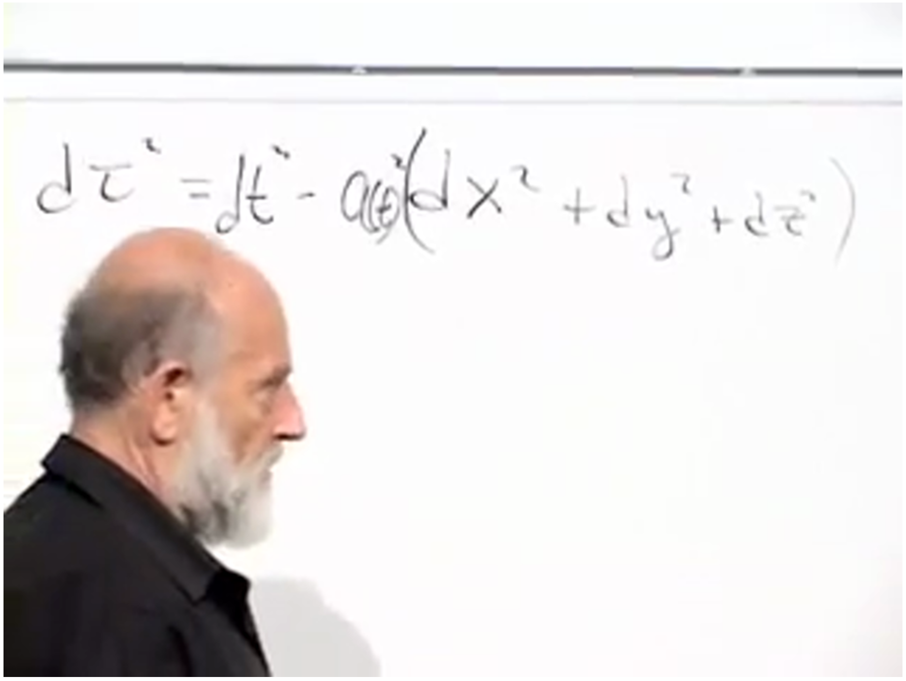

Figure 1.2: The flat expanding metric, with the scale factor multiplying the Euclidean spatial line element. Source frame at 00:09:12.

Imagine coordinates painted on a rubber sheet. The coordinate marks remain attached to the sheet while the physical distance between them grows. Thus a coordinate difference such as "from $x = 0$ to $x = 1$" is only a pair of labels. The distance represented by that interval is $a(t)$. Finding the equation that governs this scale factor is the dynamical problem of the lecture.

1.2 Closed, flat, and open spatial geometries

Flatness is an observational simplification, not a consequence of homogeneity. There are three homogeneous and isotropic spatial geometries: positive, zero, and negative curvature. To understand the positive case, it helps to be precise about dimension.

The locus

$$X_1^2 + X_2^2 = R^2 \tag{1.3}$$

is a circle, or one-sphere S^1. Its filled interior is a disk, also called a two-ball. Adding one embedding coordinate gives

$$X_1^2 + X_2^2 + X_3^2 = R^2, \tag{1.4}$$

the ordinary two-sphere S^2. Adding a fourth embedding coordinate gives

$$X_1^2 + X_2^2 + X_3^2 + X_4^2 = R^2, \tag{1.5}$$

a three-sphere S^3, which is itself a three-dimensional space. We cannot picture its four-dimensional embedding, but the intrinsic geometry does not need that embedding in order to exist.

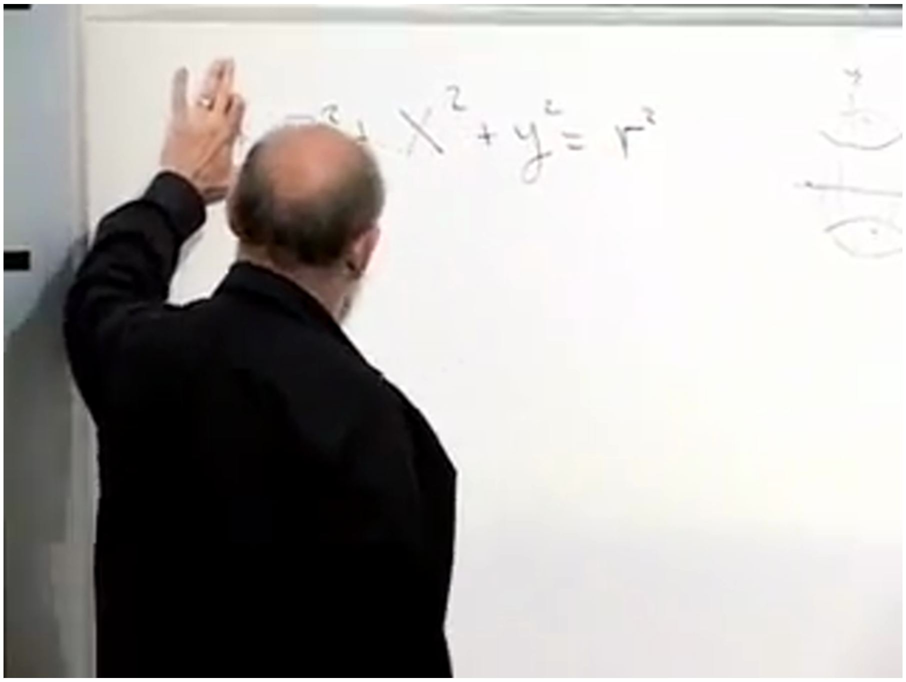

Figure 1.3: The embedding equation used to introduce the three-sphere. Source frame at 00:16:40.

Every point of an ideal sphere is geometrically equivalent to every other, and every direction at a point is equivalent. A sphere is therefore both homogeneous and isotropic. If its radius is multiplied by $a(t)$, the whole space expands or contracts without selecting a center on the sphere. Tracing such a universe backward does not place all matter at one special location in an exterior void. The entire space becomes small, so every pair of points becomes close. The Big Bang is not an explosion from one point of space.

On a positively curved sphere, geodesic triangles have angle sum greater than 180°. Flat triangles have angle sum exactly 180°. In a homogeneous negatively curved geometry, represented imperfectly by a saddle, the angle sum is less than 180°. These intrinsic tests distinguish the three cases.

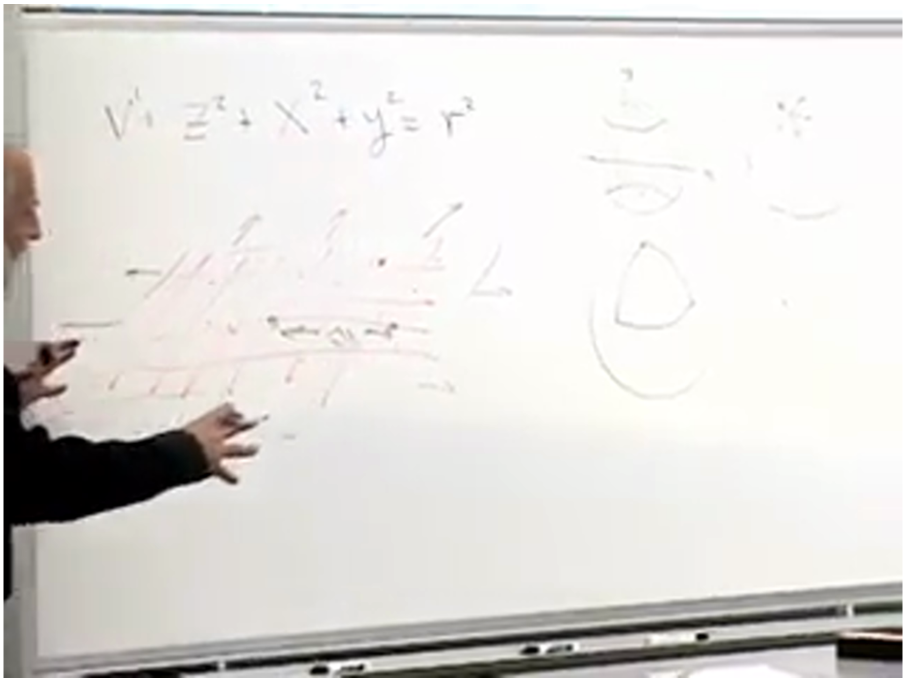

Figure 1.4: Positive, zero, and negative curvature sketched through their characteristic triangle shapes. Source frame at 00:37:30.

A perfectly flat local metric does not by itself prove that space is topologically infinite. Compact flat topologies are mathematically possible, although the lecture adopts the simplest infinite Euclidean model. [1]

1.3 Comoving labels and Hubble's law

Return now to the expanding flat sheet and draw a coordinate grid on it. In comoving coordinates, the grid expands with the large-scale matter flow. A galaxy, or more accurately a sufficiently coarse-grained cluster of galaxies, keeps an approximately fixed coordinate label even though its physical distance from another comoving object changes.

[1] Editorial clarification: This global-topology qualification is not needed for any of the subsequent local dynamics.

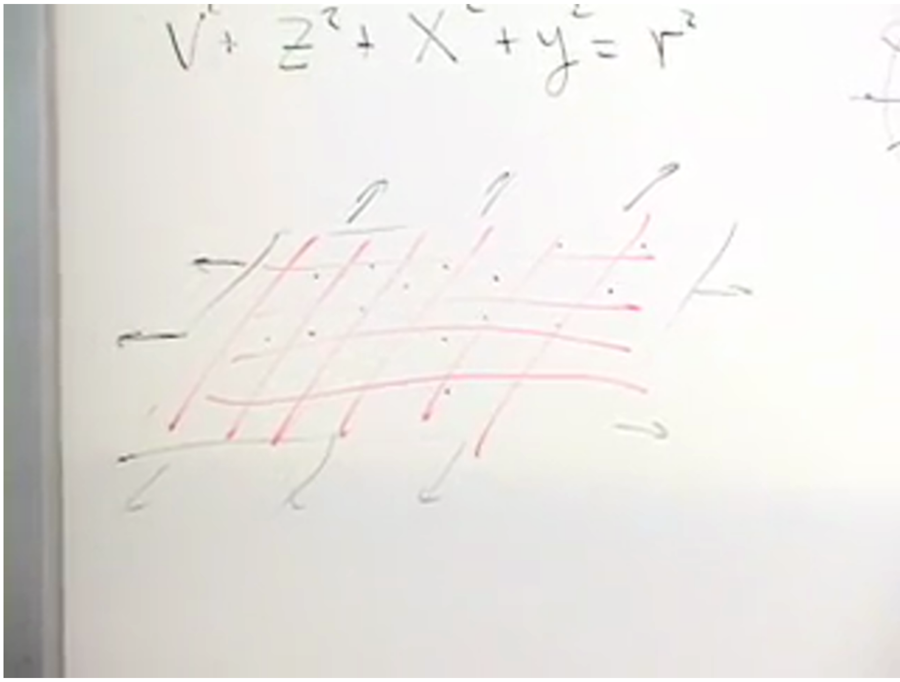

Figure 1.5: A coordinate mesh carried along by an expanding spatial sheet. Source frame at 00:22:20.

For two objects separated along the x-axis by a fixed comoving interval Δx, their physical distance is

$$D(t) = a(t)\,\Delta x. \tag{1.6}$$

Differentiating while holding Δx fixed gives

$$v(t) = \dot{a}(t)\,\Delta x = \frac{\dot{a}(t)}{a(t)}D(t). \tag{1.7}$$

Defining

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)}, \tag{1.8}$$

we obtain Hubble's law,

$$v = H(t)D. \tag{1.9}$$

The word "constant" in "Hubble constant" means constant across space at a specified cosmic time. It does not imply constancy through cosmic history.

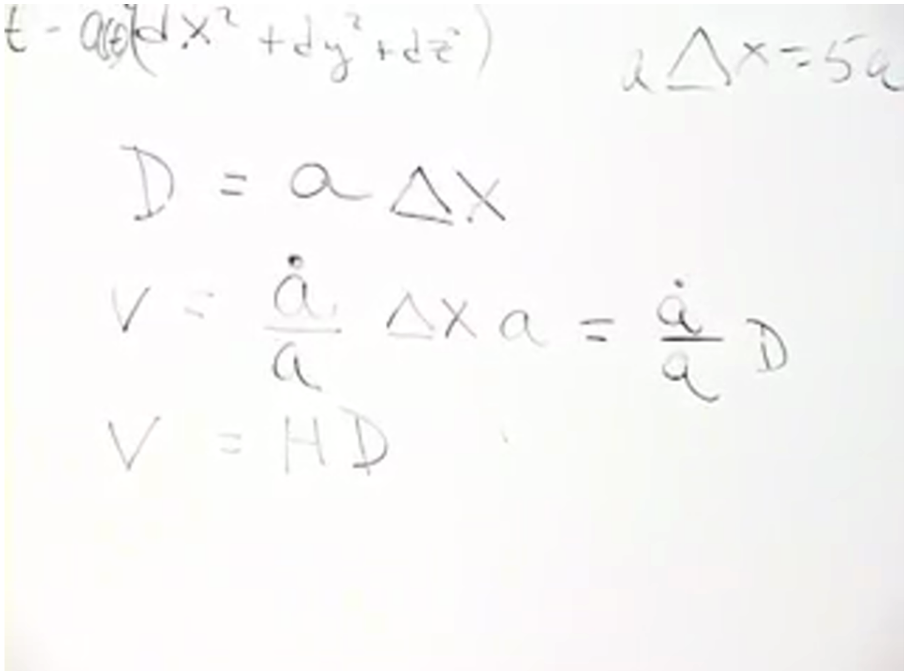

Figure 1.6: The board derivation $D = a\Delta x$, $v = (\dot{a}/a)D = HD$. Source frame at 00:26:40.

Question from the audience. If a distant galaxy is seen at an earlier time, should its inferred Hubble parameter really equal the value measured nearby today?

Response. Only when the lookback time is small compared with the age of the universe. Over modest distances, $H(t)$ changes little enough that one number is useful. At truly cosmological distances we see earlier epochs, and the time dependence of a and H must be included. That is one reason we need a dynamical equation for the scale factor.

1.4 Null rays in an expanding coordinate system

Question from the audience. Does the scale factor affect the propagation of light between distant galaxies?

Response. A light ray follows a null trajectory, so $d\tau = 0$. For a ray moving along the positive x-axis, $dy = dz = 0$, and the expanding metric gives

$$0 = dt^2 - a^2(t)dx^2, \qquad dt = a(t)dx, \qquad \frac{dx}{dt} = \frac{1}{a(t)}. \tag{1.10}$$

The coordinate speed decreases as $a(t)$ grows, but the local physical speed of light remains $c = 1$. A fixed coordinate interval represents an increasing physical distance.

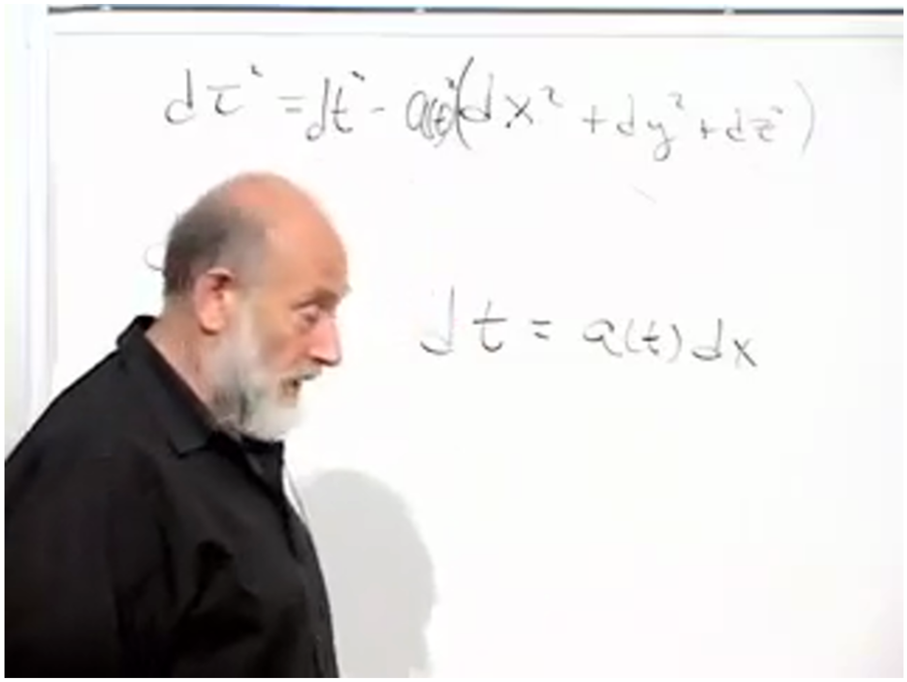

Figure 1.7: The null condition and $dt = a(t)dx$ for radial light propagation. Source frame at 00:30:54.

One can choose coordinates in which this same fact sounds like "light slows down farther away," but then lengths and clocks acquire correspondingly awkward descriptions. The expanding-space language is cleaner because it preserves the local statement that every freely falling observer measures the same speed of light.

Question from the audience. Is there evidence that local light propagation and atomic physics are the same in distant galaxies?

Response. Only dimensionless comparisons are operationally meaningful. Choose an atomic size as a unit of length and an atomic or nuclear period as a unit of time, then ask how many such periods light requires to cross the chosen length. Spectral observations support the statement that these dimensionless ratios are the same in distant regions. The evidence is indirect, but it is precisely the kind of comparison by which one tests the uniformity of local laws.

When the observable universe was extremely small, light could cross a given physical region quickly in the corresponding light-crossing unit. This does not mean that c was different. It means that the region itself was small.

1.5 A Newtonian sphere inside the universe

We now turn from kinematics to dynamics. Place ourselves at the origin and draw a sphere large enough to contain many galaxies and clusters. Select a galaxy on its boundary. In a homogeneous universe the mass distribution is spherically symmetric around us, and Newton's shell theorem gives two crucial facts:

1. all spherically distributed matter outside the chosen radius exerts zero net Newtonian force

on the boundary galaxy;

2. all matter inside the sphere acts, for this radial motion, as if it were concentrated at the center.

We use a frame in which we are at rest, so velocities and accelerations in the following argument are measured relative to us.

Figure 1.8: The shell-theorem construction: us at the center and a test galaxy on the boundary of a large homogeneous sphere. Source frame at 00:41:10.

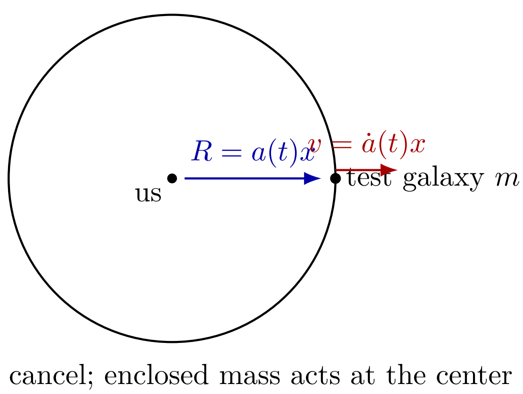

Figure 1.9: Clean reconstruction of the Newtonian construction.

If the galaxy has fixed comoving coordinate x, its physical radius and velocity are

$$R(t) = xa(t), \qquad v(t) = x\dot{a}(t) = H(t)R(t). \tag{1.11}$$

Let M be the mass enclosed by the sphere and m the mass of the test galaxy. The Newtonian energy is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}. \tag{1.12}$$

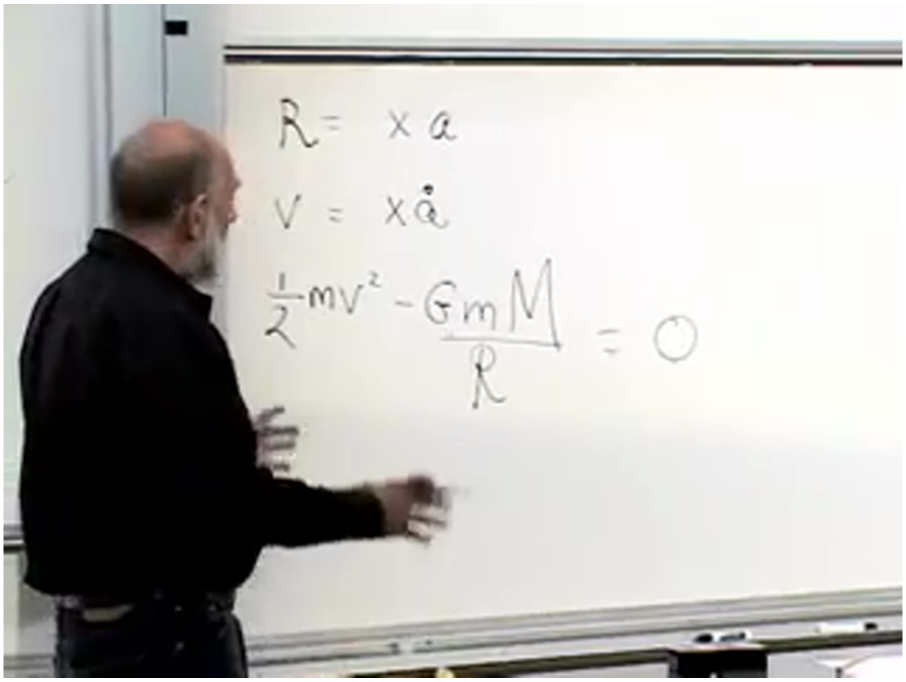

Figure 1.10: The complete board sequence $R = xa$, $v = x\dot{a}$, and the Newtonian energy equation specialized to $E = 0$. Source frame at 00:48:40.

The energy may be positive, zero, or negative. Start at the boundary between escape and recollapse, $E = 0$, for which

$$\frac{1}{2}v^2 = \frac{GM}{R}. \tag{1.13}$$

Question from the audience. What does zero total energy mean for the receding galaxy?

Response. It is exactly the escape-velocity case. Positive energy means motion above escape velocity; negative energy means motion below escape velocity and, in the matter-only Newtonian picture, eventual return. We begin at the boundary between those possibilities and return to the other signs after deriving the flat equation.

Question from the audience. Does the enclosed mass M vary as the sphere expands?

Response. No. The sphere is comoving, so the same particles remain inside it. Its physical volume grows and its density falls, but its total enclosed mass remains fixed. This distinction between M and $\rho(t)$ is essential.

For a homogeneous density $\rho(t)$,

$$M = \frac{4\pi}{3}\rho(t)R^3. \tag{1.14}$$

Substitution into (1.13) gives

$$\frac{1}{2}v^2 = \frac{G}{R}\left(\frac{4\pi}{3}\rho R^3\right). \tag{1.15}$$

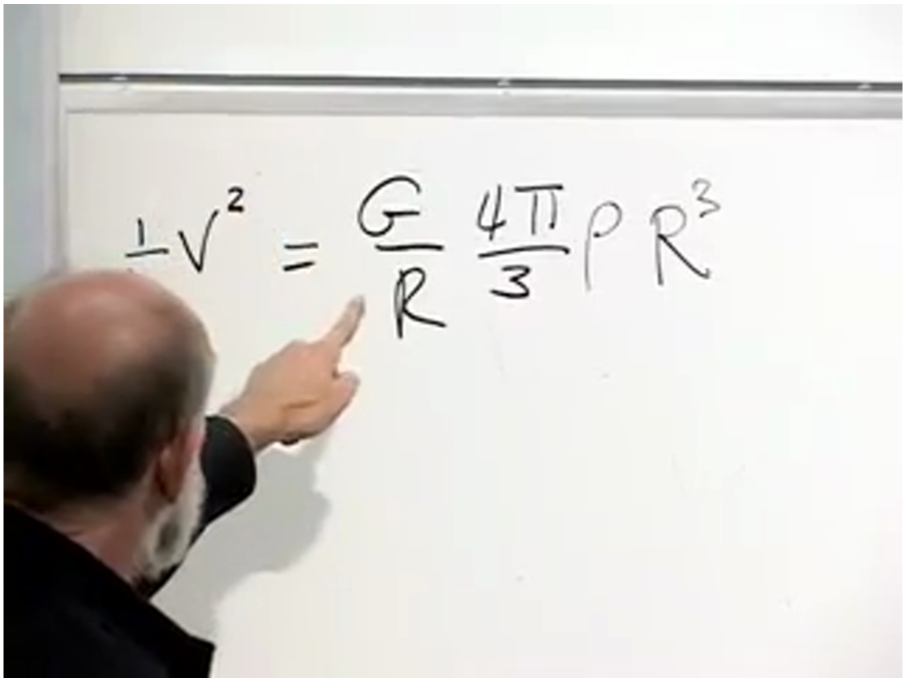

Figure 1.11: The enclosed-mass substitution in the zero-energy equation. Source frame at 00:54:16.060.

Using $v = HR$, we find

$$\frac{1}{2}H^2R^2 = \frac{4\pi G}{3}\rho R^2, \tag{1.16}$$

$$H^2 = \frac{8\pi G}{3}\rho. \tag{1.17}$$

Thus the zero-energy, spatially flat expansion law is

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho(t)}. \tag{1.18}$$

The cancellation of R^2 is not accidental. A homogeneous universe must give the same $H(t)$ no matter which comoving galaxy was chosen as the test object.

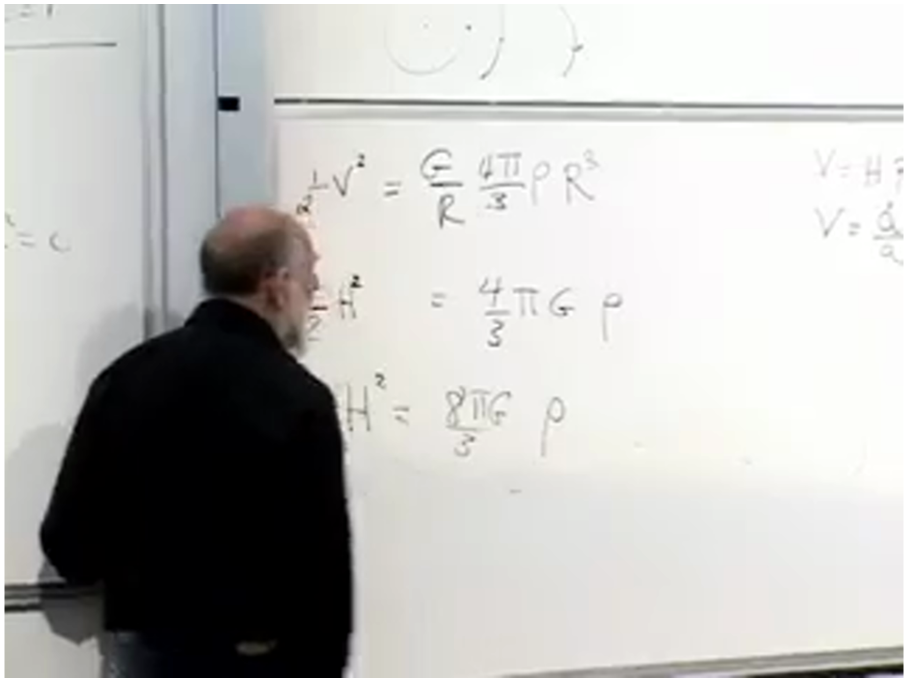

Figure 1.12: The cancellation of the radius and the resulting flat Friedmann relation on the board. Source frame at 00:57:30.

Question from the audience. Is there evidence for the zero-energy relation?

Response. Both sides can be inferred observationally: H from recession data and ρ by accounting for stars, gas, dark matter, and vacuum energy. The lecture-era comparison found the two sides consistent to about percent accuracy. The qualification matters: all forms of energy must be included, and the later vacuum-energy discussion changes the naive escape-velocity interpretation.

In units $c = 1$, mass density and energy density have the same units. The symbol ρ will therefore denote the total energy density appropriate to the component under discussion.

1.6 Ordinary matter and the two-thirds law

Equation (1.18) is not closed until we know how $\rho(t)$ changes. First fill a comoving rectangular box with slowly moving, nonrelativistic particles. Its coordinate dimensions are fixed, while its physical volume grows as

$$V(t) = a^3(t)\,\Delta x\,\Delta y\,\Delta z. \tag{1.19}$$

The particle number and rest mass in the box remain fixed, so

$$\rho_{\mathrm{m}}(t) = \frac{C_{\mathrm{m}}}{a^3(t)}. \tag{1.20}$$

Normalizing to the present epoch t_0,

$$\frac{\rho_{\mathrm{m}}(t)}{\rho_{\mathrm{m},0}} = \left(\frac{a_0}{a(t)}\right)^3. \tag{1.21}$$

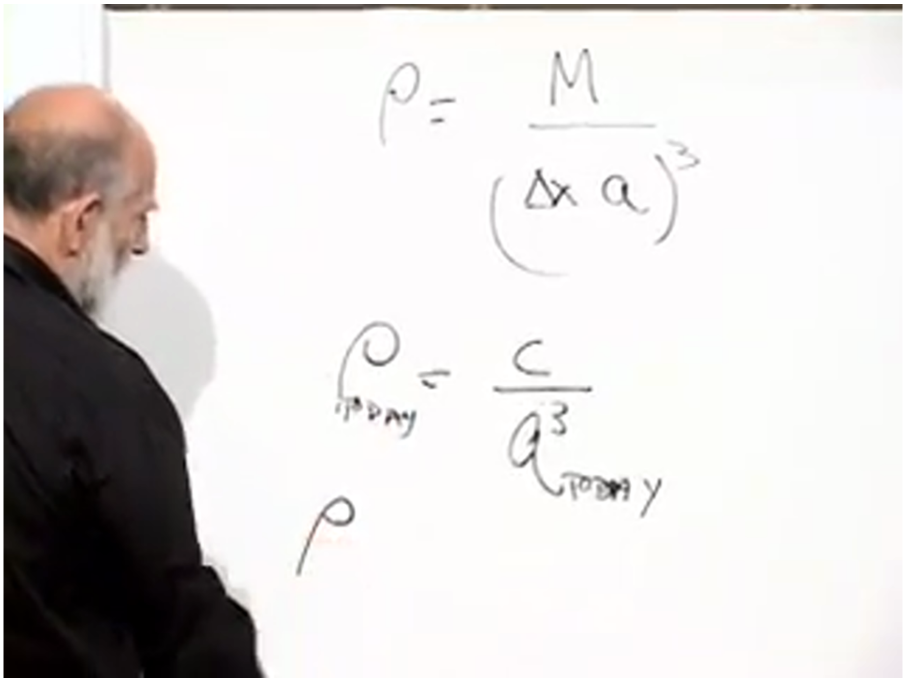

Figure 1.13: Matter density, its a^{-3} dilution, and present-day normalization. Source frame at 01:04:47.940.

Question from the audience. What is the rough present energy density of the universe, and can ordinary atoms alone provide it?

Response. A count of visible baryonic matter gives only part of the answer. Dark matter and dark energy must also be represented as equivalent mass-energy. The lecture quotes a rough scale ranging from an atom to a few dozen hydrogen-mass equivalents per cubic meter, emphasizing order of magnitude rather than a precision inventory.

Insert (1.20) into (1.18) and absorb all time-independent factors into a positive constant C:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{C}{a^3}, \qquad \dot{a}^2 = \frac{C}{a}. \tag{1.22}$$

There are two transparent ways to solve it. A power-law trial $a = Kt^p$ requires

$$p - 1 = -\frac{p}{2}, \qquad p = \frac{2}{3}. \tag{1.23}$$

Equivalently, take the expanding square-root branch and integrate:

$$a^{1/2}\frac{da}{dt} = \sqrt{C}, \qquad \int a^{1/2} da = \sqrt{C}\int dt, \tag{1.24}$$

so, after choosing the Big Bang origin of time,

$$a^{3/2} \propto t, \qquad \boxed{a(t) \propto t^{2/3}}. \tag{1.25}$$

The live derivation briefly loses a factor while matching coefficients, then returns to the integral and verifies the exponent. The notes retain the check but display only the corrected algebra.

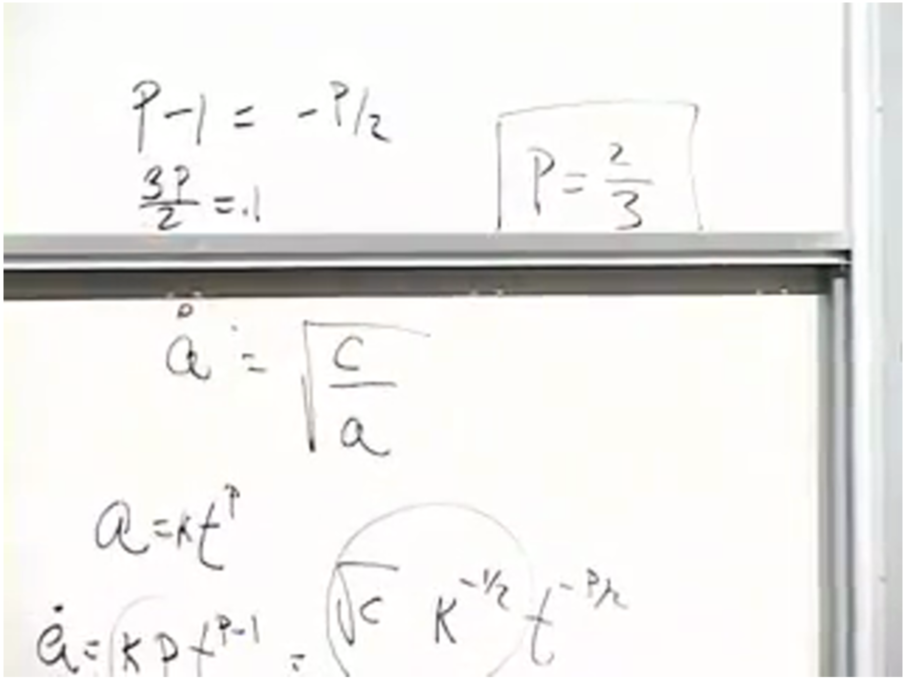

Figure 1.14: The power-law trial and the boxed exponent $p = 2/3$. Source frame at 01:13:05.

Differentiating the solution gives

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{1.26}$$

The Hubble parameter was larger in the past. Consequently, when looking far enough away to see appreciable lookback time, one should not extend the local linear relation using today's H_0 unchanged.

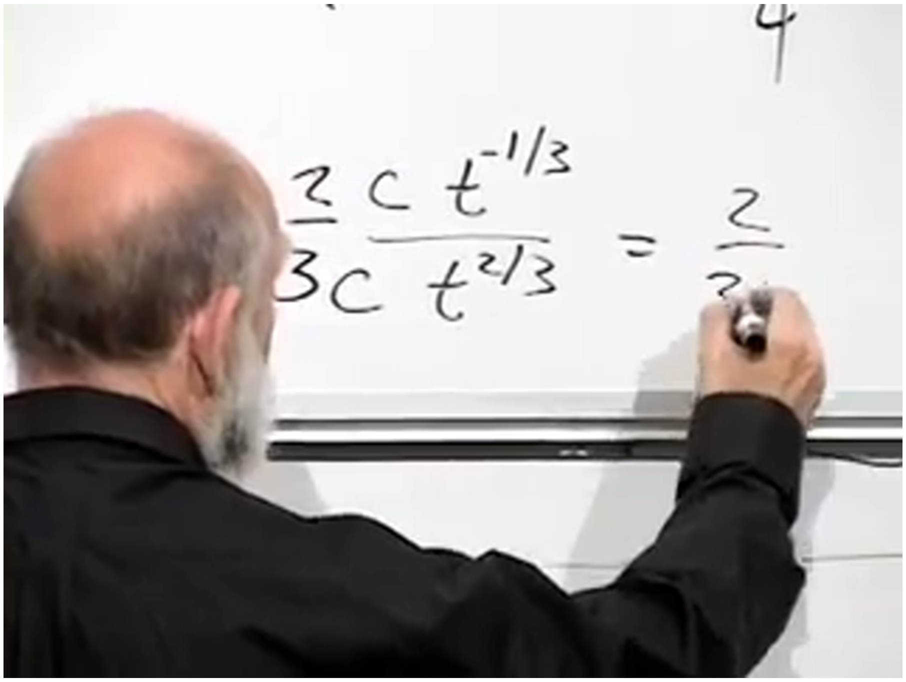

Figure 1.15: Taking $\dot{a}/a$ for $a \propto t^{2/3}$ gives $H = 2/(3t)$. Source frame at 01:20:42.

1.7 Energy sign and spatial curvature

Newtonian reasoning has carried us through the flat zero-energy case. General relativity adds a fact that cannot be obtained from the shell theorem alone: the sign of the Newtonian-style specific energy corresponds to the sign of the spatial curvature.

Total energy	Spatial curvature	Matter-only fate
positive	negative (open)	expands forever
zero	zero (flat)	approaches escape boundary
negative	positive (closed)	expands, turns, recollapses

The last column assumes no vacuum energy. In the projectile analogy, positive energy means above escape velocity, zero means exactly at escape velocity, and negative means below escape velocity. Einstein's equations identify these three cases with open, flat, and closed spatial geometry respectively.

1.8 Photons and the one-half law

The matter law $\rho \propto a^{-3}$ assumes that the energy of each particle is approximately fixed while the number density falls. Photons violate that assumption. For a photon,

$$E = h\nu = \frac{hc}{\lambda}. \tag{1.27}$$

Expansion stretches its wavelength in proportion to the scale factor,

$$\lambda \propto a, \qquad E \propto a^{-1}. \tag{1.28}$$

The photon number density still falls as a^{-3}, and each photon's energy contributes one additional factor of a^{-1}. Therefore

$$\boxed{\rho_{\mathrm{r}}(t) \propto a^{-4}(t)}. \tag{1.29}$$

The guitar-string analogy captures the physical picture. Pluck a string and then lengthen its vibrating portion: the wavelength lengthens and the tone drops. A photon mode in an expanding comoving box behaves similarly. Running the picture backward, compression shortens wavelengths and raises photon energies.

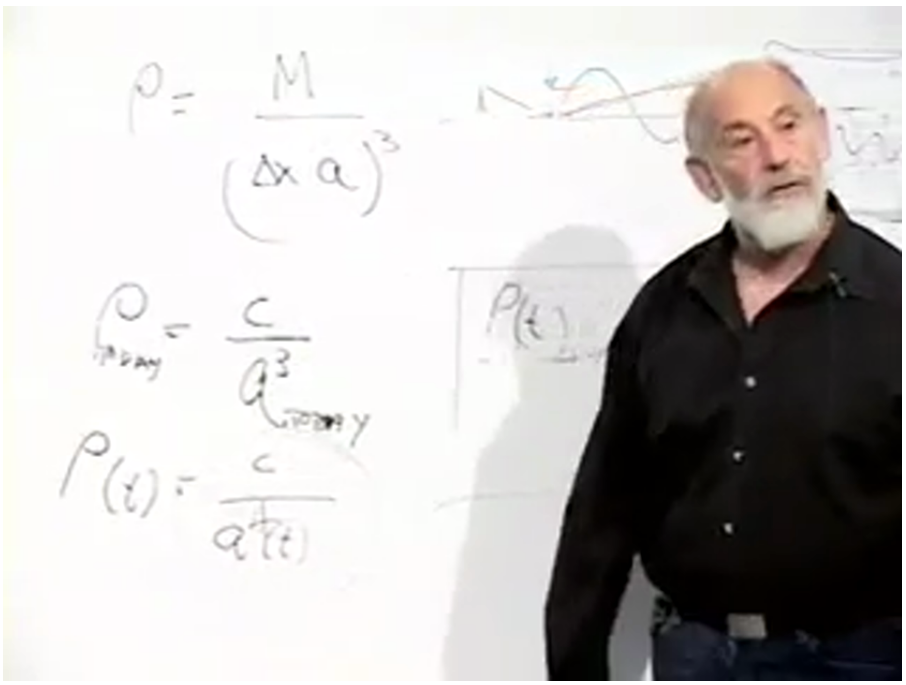

Figure 1.16: The matter-density relation extended on the board to the radiation scaling $\rho_{\mathrm{r}} \propto a^{-4}$. Source frame at 01:29:20.

Using $\rho_{\mathrm{r}} \propto a^{-4}$ in the flat Friedmann equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{C_{\mathrm{r}}}{a^4}, \qquad \dot{a} \propto a^{-1}, \tag{1.30}$$

and hence

$$\boxed{a(t) \propto t^{1/2}}. \tag{1.31}$$

This grows more slowly than $t^{2/3}$. Far enough into the past, radiation must dominate ordinary matter because a^{-4} rises faster than a^{-3} as a decreases. The lecture places the matter-radiation crossover around the epoch when the scale factor was roughly a thousand times smaller than today. Before that crossover the universe was radiation dominated; after it, for a long interval, it was matter dominated.

1.9 Vacuum energy and exponential expansion

Vacuum energy has the opposite scaling behavior. Its density is a property of empty space and does not dilute as the universe expands:

$$\rho_\Lambda = \text{constant}. \tag{1.32}$$

Ordinary matter and radiation become denser toward the past, while this term does not. It was therefore less important in the early universe and becomes more important as the other components dilute. The lecture uses the then-current estimate that vacuum energy supplies about 70% of today's cosmic energy density.

With constant ρ_Λ, equation (1.18) gives a constant expansion rate

$$H_\Lambda = \sqrt{\frac{8\pi G}{3}\rho_\Lambda}, \qquad \frac{\dot{a}}{a} = H_\Lambda. \tag{1.33}$$

Therefore

$$\boxed{a(t) = a(t_i)e^{H_\Lambda(t-t_i)}}. \tag{1.34}$$

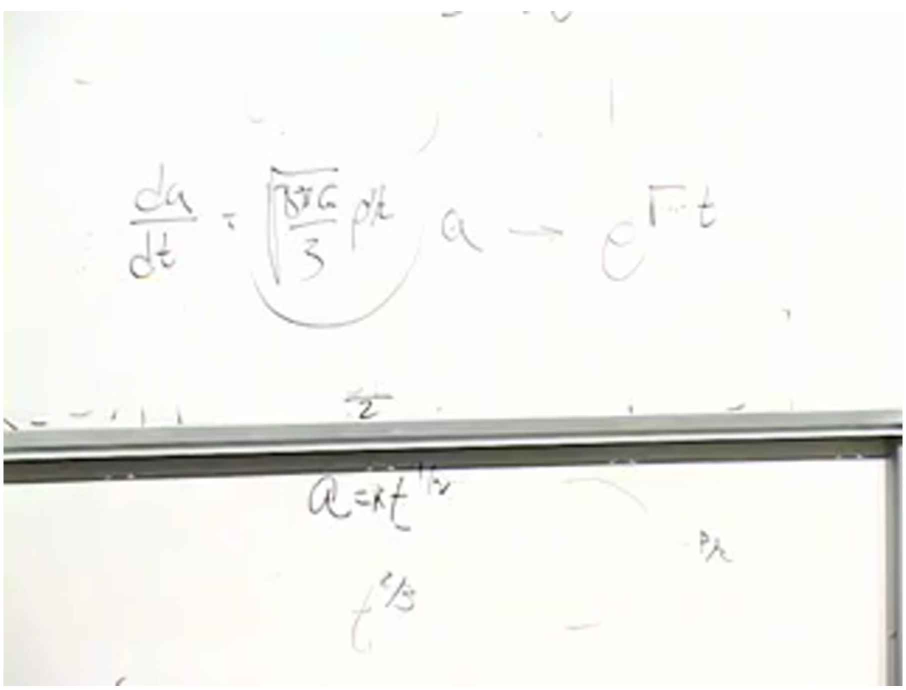

Figure 1.17: The constant-density equation leading to exponential expansion. Source frame at 01:35:48.

At the observed scale quoted in the lecture, the universe doubles in linear size over roughly ten to twelve billion years. The origin of this small but nonzero vacuum density remains unexplained; its existence and its effect on the expansion are empirical facts.

Question from the audience. Is vacuum energy the same thing as dark energy in this discussion?

Response. Yes. Here "dark energy" denotes a constant vacuum-energy density, equivalently a cosmological constant. More general time-dependent dark-energy models can be imagined, but they are not part of this lecture's calculation.

1.10 How an expanding universe redshifts light

Question from the audience. How can photons know that we chose expanding comoving coordinates and therefore lower their energy?

Response. They do not respond to our labels. They propagate through a geometry whose physical separations change. A wave train occupying a region that doubles in physical size acquires a doubled wavelength. The comoving coordinates merely describe this stretching economically; they do not cause it.

Question from the audience. Where do the background photons reaching us today come from?

Response. Looking farther away means looking earlier. Eventually we see the epoch when the cosmic material was an ionized plasma, opaque because photons scattered repeatedly from free charged particles. The boundary from which photons could last travel freely to us appears as a sphere surrounding the observer: the surface of last scattering.

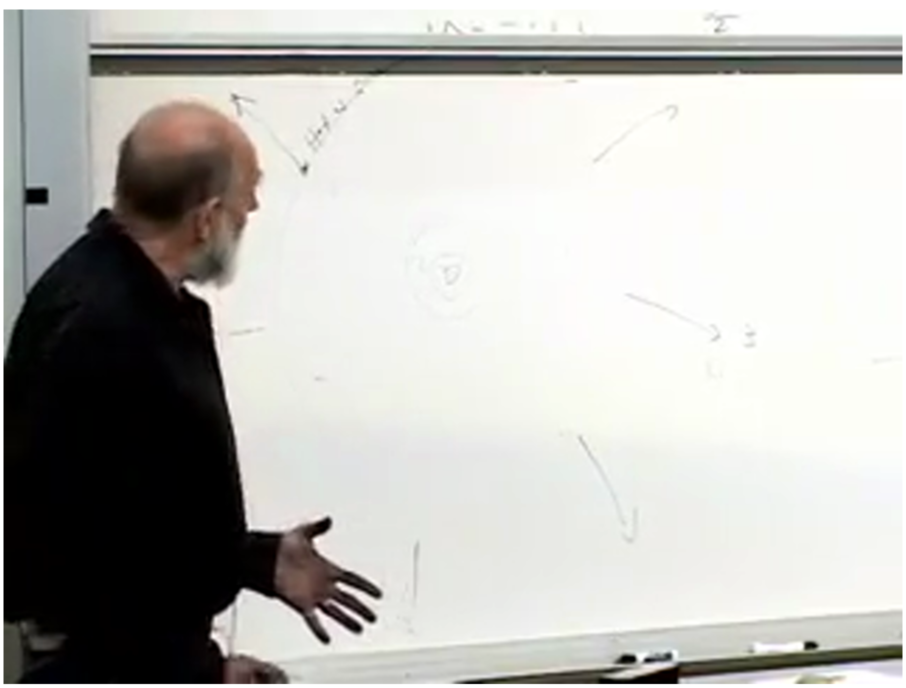

Figure 1.18: The observer-centered sketch of the surface of last scattering and increasingly distant shells. Source frame at 01:42:05.

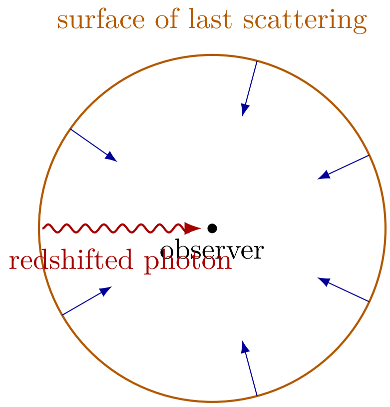

Figure 1.19: Clean reconstruction of the last-scattering picture.

The radiation was emitted with a spectrum characteristic of a hot plasma and reaches us with wavelengths roughly a thousand times longer. In the lecture's Doppler language, the distant emitting surface recedes rapidly and its radiation is strongly redshifted. In the geometric language, the wave is stretched during propagation. Both pictures encode the same relation,

$$1 + z = \frac{\lambda_0}{\lambda_{\text{em}}} = \frac{a_0}{a_{\text{em}}}. \tag{1.35}$$

[2]

At an earlier observing epoch, the last-scattering sphere would have been closer in the corresponding picture, and the observed background photons would have been less redshifted and therefore more energetic. As cosmic time advances, photons reaching a comoving observer from that early plasma have progressively longer wavelengths.

1.11 Where does a redshifted photon's energy go?

Question from the audience. A photon is emitted with energy $h\nu_{\text{em}}$ and arrives redshifted with less energy. What happened to the difference?

Response. The Newtonian derivation began as an energy equation. Its cosmological form relates the expansion rate to the material energy density, so matter energy and expansion cannot be treated as separately conserved accounts. The lecture describes the lost photon energy as entering the kinetic bookkeeping of expansion. A piston filled with radiation makes the analogy concrete: radiation pressure pushes the piston outward, the photon wavelengths increase, and their lost energy becomes mechanical work and kinetic energy of the piston.

[2] Editorial clarification: Equation (1.35) is the standard quantitative form of the lecture's stretching and Doppler descriptions.

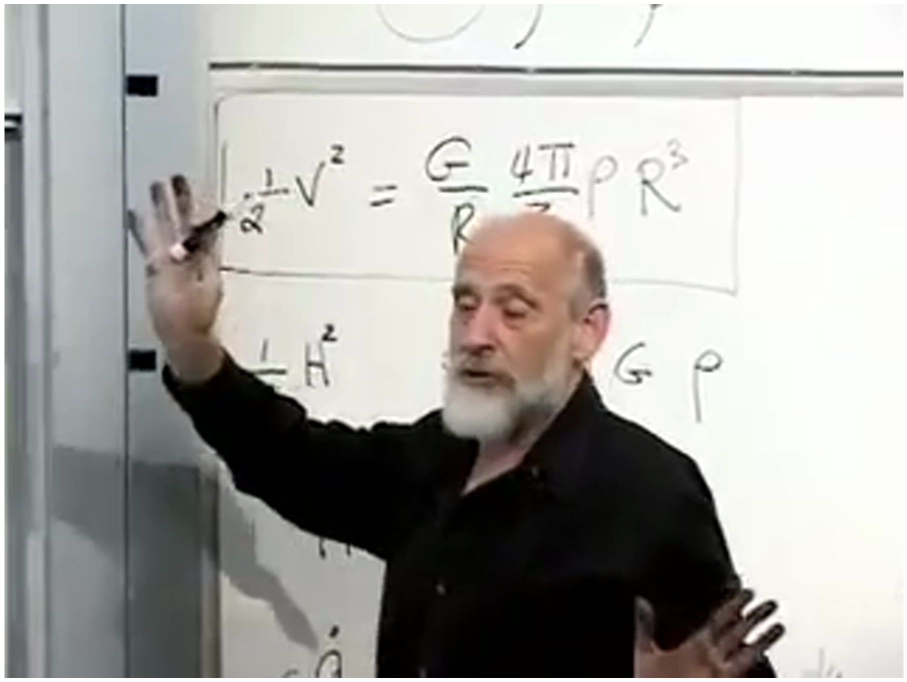

Figure 1.20: The boxed Newtonian energy equation above its Friedmann form during the energy-conservation discussion. Source frame at 01:44:42.440.

Question from the audience. What couples the radiation energy to the expansion?

Response. Pressure. In a cylinder with a movable piston, photons transfer momentum to the wall. As the piston moves, the radiation does work; its wavelength grows and its energy falls. The homogeneous expanding universe has no literal outer piston, but the $p\,dV$ analogy explains why the energy density of radiation falls faster than number dilution alone.

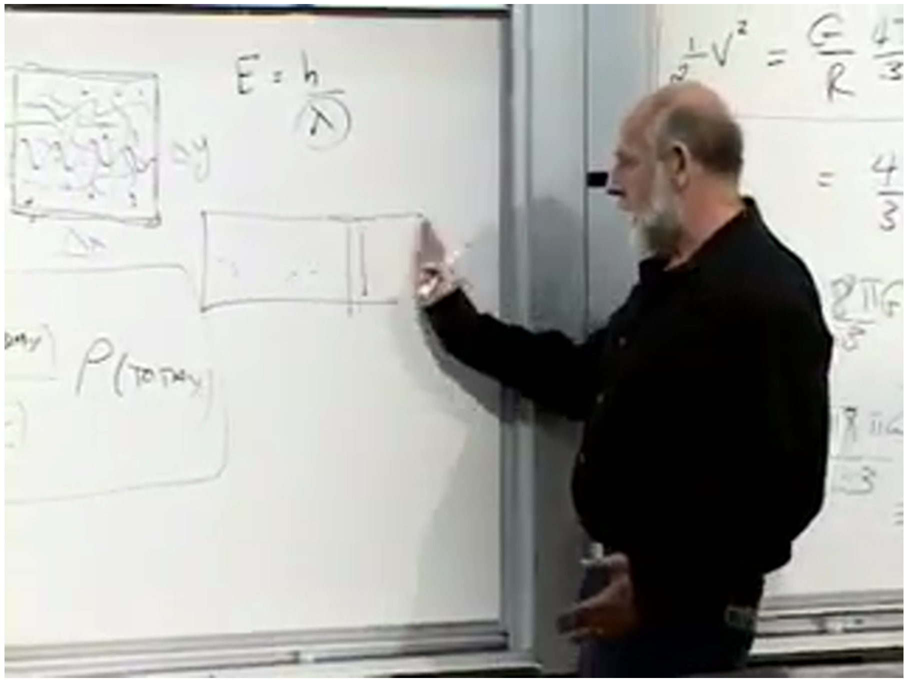

Figure 1.21: The movable photon-filled piston used to visualize pressure work. Source frame at 01:46:40.

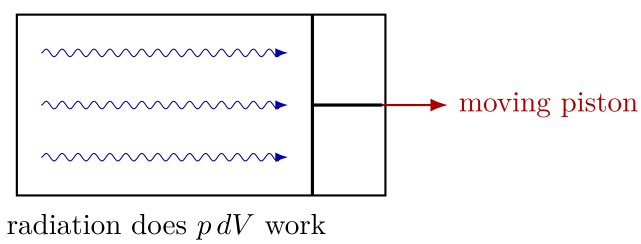

Figure 1.22: Clean reconstruction of the radiation-piston analogy.

Energy in general relativity requires care. In a generic expanding spacetime there need not be a unique conserved global energy analogous to the energy of an isolated Newtonian system. Local stress-energy conservation still holds, and for a homogeneous fluid it yields the continuity equation

$$\dot{\rho} + 3H(\rho + p) = 0, \tag{1.36}$$

which contains the pressure-work term behind the matter and radiation scaling laws.[3]

Vacuum energy makes the puzzle especially sharp. Since its density stays constant while volume grows, the total vacuum energy in a comoving region increases. Its negative pressure is part of the gravitational source and drives accelerated expansion. One should therefore not demand separate conservation of the naive "matter energy in a comoving box" without including the geometry and pressure work.

[3]Editorial clarification: Equation (1.36) is the precise local general-relativistic statement underlying the lecture's deliberately Newtonian energy-bookkeeping analogy.

1.12 Why the scale factor can be removed from the time term

Question from the audience. Why does the scale factor multiply the spatial part of the metric but not the time part?

Response. That is a choice of time coordinate. Suppose one begins in a conformal-time coordinate t with

$$d\tau^2 = A^2(t)\left(dt^2 - dx^2\right). \tag{1.37}$$

Define a new cosmic time T by

$$dT = A(t)\,dt. \tag{1.38}$$

Then

$$d\tau^2 = dT^2 - A^2(t(T))dx^2. \tag{1.39}$$

The common factor in front of the time differential has disappeared, while the physical scale factor remains in the spatial term.

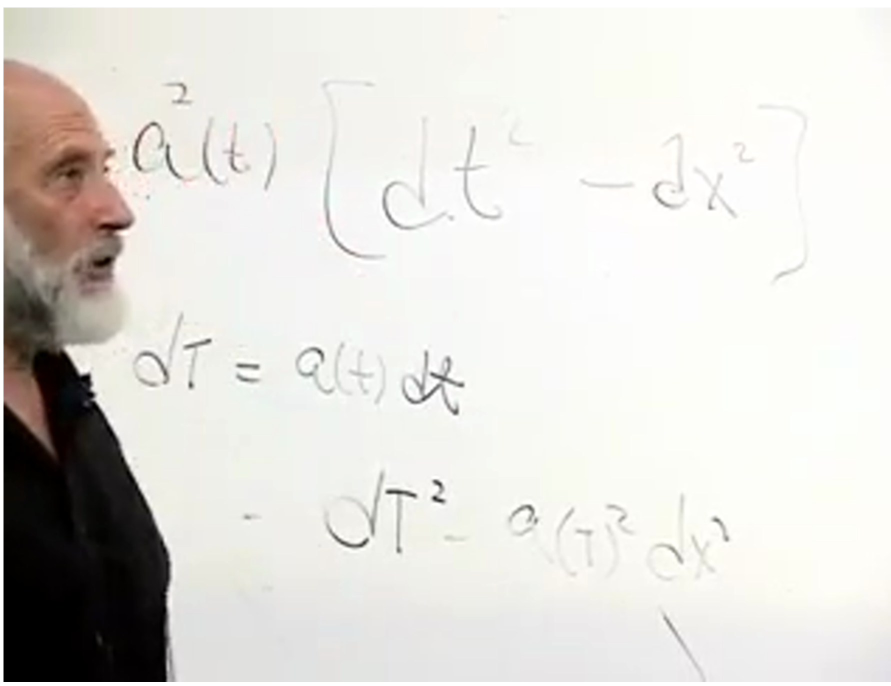

Figure 1.23: The time redefinition $dT = A(t)dt$ that converts conformal time to cosmic time. Source frame at 01:51:05.

1.13 Curvature, vacuum energy, and cosmic fate

Question from the audience. If the universe were positively curved, should we already see galaxies coming toward us?

Response. Not necessarily. A projectile launched below escape velocity first travels outward, reaches a turning point, and only later falls back. We observe the universe during an expanding phase. Without vacuum energy, a positively curved matter universe would eventually recollapse. A sufficiently large positive vacuum energy prevents that turnaround, and the observed value appears large enough to make the long-term expansion exponential regardless of a small spatial curvature.

The exponential future is not an energetic paradise. Stars eventually burn out, matter and radiation densities fall exponentially, and distant regions pass beyond causal contact. Vacuum energy remains while ordinary structure becomes progressively colder and more dilute.

Question from the audience. Looking farther into the past, are there more photons than ordinary particles?

Response. The photon-to-baryon number ratio is already enormous, roughly 10^{10} photons per baryon in the lecture's estimate, and both numbers in a comoving volume remain approximately fixed. What changes toward the past is mainly the energy per photon. Photon energies rise as a^{-1}, while nonrelativistic rest masses do not. Radiation therefore dominates the energy density at early times even though the number ratio itself need not change.

The central lesson is that geometry alone does not determine the cosmic history. The flat Friedmann equation supplies the relation between expansion and energy density, but the material content supplies the time dependence. Nonrelativistic matter gives $a \propto t^{2/3}$, radiation gives $a \propto t^{1/2}$, and constant vacuum energy gives exponential expansion. Those three laws organize the successive eras of the universe and explain why the same scale factor controls distances, redshifts, and the changing Hubble parameter.